

BACTERIAL RESISTANCE TO ANTIMICROBIAL AGENTS IN LATIN AMERICA

The Giant Is Awakening

Manuel Guzmán-Blanco, MD, FACP, José María Casellas, MB,
and Helio Silva Sader, MD

Resistant bacteria are emerging in Latin America as a real threat to the favorable outcome of common infections in community and hospital settings.

The time to change long-accepted prescription practices is coming, and every effort should be made to provide the prescribers the pertinent information related to changing patterns of resistance.

Adequate and prompt surveillance of resistance, identification of the factors that make possible the spread of the mechanisms of resistance, and recommendations for the adequate usage of antibiotics have been recently suggested for the region.⁵²

This article updates the situation as of mid-1999. In a previous issue of this journal, results up until 1993 were presented.⁷

SURVEILLANCE OF RESISTANCE IN LATIN AMERICA

Surveillance of resistance is currently very active in Latin America (Table 1). Programs sponsored by multinational organizations (World

From the Infectious Disease and Microbiology Unit, Hospital Vargas de Caracas and Centro Médico de Caracas, Caracas, Venezuela (MG-B); Infectious Disease and Clinical Microbiology Unit, San Lucas Hospital, San Isidro, Buenos Aires, Argentina (JMC); and the Division of Infectious Diseases, Federal University of São Paulo, São Paulo, Brazil (HSS)

Table 1. PROGRAMS OF SURVEILLANCE OF RESISTANCE CURRENTLY ACTIVE IN LATIN AMERICA

WHO Sponsored: Antimicrobial Resistance Network WHONET
PAHO "problem organisms": <i>Streptococcus pneumoniae</i>
Intestinal pathogens
Gonococci resistance network
Pan American Association for Infectious Diseases (API) surveys
National Societies Programs
Pharmaceutical Companies: Bristol Myers Squibb: SENTRY
Merck & Co.: INR
Pfizer: ARTEMIS/RESISNET
Rhone Poulenc: GSMART

Health Organization [WHO], Pan American Health Organization [PAHO], Panamerican Society of Infectious Diseases), national surveillance programs, and information gathered by pharmaceutical company-sponsored studies have provided pertinent and current information about the situation in the area. In the conclusions of a recent meeting sponsored by the PAHO in Caraballeda, Venezuela, the need for a Latin American surveillance system available on the internet was highly recommended.⁵²

The WHO supports WHONET, a computer program and a network of microbiologists that uniformly collects and analyzes antimicrobial resistance data as part of global surveillance.⁵⁴ Reports from Venezuela were recently published.⁴⁴ Argentina, Uruguay, and Colombia are also part of the Antimicrobial Resistance Monitoring System, which recently included a quality-control and proficiency testing program. Early reports from this system suggest that the disk diffusion susceptibility testing method provides accurate results, and most laboratories are providing reliable data to the WHONET system.⁵⁸

The PAHO sponsors several surveillance programs based on problem organisms. Results of resistance patterns of invasive strains of *Streptococcus pneumoniae* in six Latin American countries were recently published.²⁴ There is a currently active program on resistance patterns in intestinal pathogens (*Salmonella*, *Shigella*, and *Vibrio*), and results are due at the end of 1999. Technical support is provided by the Center for Disease Control of Canada (LDCDC). The Gonococcal Antimicrobial Surveillance System is also present in several countries of the region.

The Panamerican Society for Infectious Diseases (API) performed two surveys on resistance in community and hospital-acquired infections, and results have been reported.^{7, 38}

Argentina, Chile, Colombia, Cuba, Ecuador, Mexico, Uruguay, and Venezuela conduct national surveillance of resistance programs, and results have been recently published.^{12, 23, 28, 46, 53, 57, 59} The Argentinian Society for Microbiology (SADEBAC) has had an active surveillance program of resistance since 1983, and the Venezuelan Society for Microbiology, since 1988.

Several pharmaceutical companies are related to surveillance sys-

tems. Results from SENTRY, sponsored by Bristol Myers Squibb,^{40, 47} and ARTEMIS from Pfizer³⁷ provide information on different aspects of resistance in the area. Other projects are currently active (RESISNET, from Pfizer, GSMART, from Rhone Poulenc; INR, from Merck & Co.).

Even though very sophisticated systems of surveillance of resistance are currently active, most of the prescribers are not aware of the results, and the necessary changes in prescription practices do not take place. Efforts to bridge this information gap have been recently suggested.⁵²

SELECTED EXAMPLES OF RESISTANT ORGANISMS

Neisseria meningitidis

Four strains that were moderately susceptible to penicillin were found among 54 consecutive isolates of meningococci recovered from patients in one pediatric hospital in Argentina from October 1991 to December 1992; one was serotype C.²⁹ In a later report,⁶ 19 of 31 serogroup meningococci recovered from adult patients with meningitis from another hospital in Buenos Aires required minimum inhibitory concentrations (MIC) of penicillin equal or greater to 0.1 µg/mL.

In a report from Caracas, Venezuela, 11 out of 11 strains of *N. meningitidis* isolated from the cerebrospinal fluid (CSF) of patients with meningeal disease between January and April of 1994 displayed MICs of penicillin G of between 0.12 and 0.25 µg/mL.²⁶ Since then, sporadic cases of meningeal disease have been mainly caused by type C meningococci with diminished susceptibility to penicillin (12 of 13 identified at Centro Médico de Caracas, between August of 1994 and July of 1996). One small outbreak in a military setting with the same type of strains was identified in September, 1997, in Caracas, and most recently (June of 1999) at a naval base in the eastern part of the country.

Twenty-one (8.5%) of 249 isolates identified in Colombia between 1987 and 1997 had diminished susceptibility to penicillin.⁴⁶ Resistance to penicillin in strains of *N. meningitidis* type *b* in Cuba increased from 16.35% in 1986 to 56% in 1997.²⁸

No report of β-lactamase producing meningococci has been made in Latin America.

Resistance in Organisms Producing Community-Acquired Respiratory Tract Infections

Acute respiratory tract infection is a major cause of childhood death in the developing world. Most respiratory tract infection-related deaths are caused by pneumonia, of which pneumococci is the most frequent bacterial cause. Other important causes of respiratory tract infections in both children and adults include *Haemophilus influenzae* and *Moraxella*

catarrhalis. Recent data describe the antimicrobial susceptibility of these bacterial pathogens in Latin America.^{24, 37}

The study conducted by the PAHO analyzed 1635 pneumococcal isolates and found that 75.1% were susceptible to penicillin ($\text{MIC} \leq 0.06 \mu\text{g/mL}$). Among the nonsusceptible strains (408 isolates), 274 (16.7% of the total) had an intermediate MIC (0.06 to $1.0 \mu\text{g/mL}$), and 134 (8.2%) exhibited high-level resistance (HLR, $\text{MIC} \geq 2 \mu\text{g/mL}$); however, there was great variation among the countries. The highest rate of nonsusceptible strains was found in Mexico (47.3%) and the lowest in Colombia (12.1%). Mexico also presented the highest rate of high-level resistance (21.3%), whereas the lowest rate was observed in Brazil (1.4%).²⁴

From January of 1997 to June of 1998, 119 (18.5%) of 643 *Streptococcus pneumoniae* identified in 10 Latin American countries were not susceptible to penicillin, 53 of them (8.2%) with a MIC of $2 \mu\text{g/mL}$ or higher.³⁷ In this study, resistance varied from 3% in Brazil to 42% in Chile; 16% of *Haemophilus influenzae* and 76% of *Moraxella catarrhalis* were β -lactamase producers.

Data from SENTRY gathered between April and September of 1998 revealed 234 strains of *S. pneumoniae*, 242 strains of *H. influenzae*, and 96 strains of *M. catarrhalis* from patients with community-acquired respiratory tract infections. The antimicrobial activity of 12 parenteral and orally administered drugs against these community-acquired pathogens is shown in Table 2. Among the 234 outpatient isolates of *S. pneumoniae* obtained, only 71.4% were susceptible to penicillin ($\text{MIC} \leq 0.06 \mu\text{g/mL}$), and 10.7% had high-level resistance ($\text{MIC} \geq 2 \mu\text{g/mL}$). Most β -lactams, including cefepime, cefotaxime, amoxicillin/clavulanate, and amoxicillin (excluding cefaclor and cefpodoxime, where the susceptibility is determined by the penicillin test results)³⁵ were active against more than 85% of strains of *S. pneumoniae*. The most active compounds against these pneumococci were levofloxacin and gatifloxacin. These newer fluoroquinolones were the only compounds active against 100% of the respiratory pathogens evaluated. Chloramphenicol was the nonfluoroquinolone compound most active against pneumococci (94.0% susceptibility). The macrolide azithromycin was active against 87.2% of strains, whereas only 52% of isolates were susceptible to trimethoprim-sulfamethoxazole (TMP-SMZ) (see Table 2).

Most of the agents tested had good activity against *H. influenzae*, with notable exceptions of TMP-SMZ (60.3% susceptible). A comparison of the susceptibility of *H. influenzae* to amoxicillin and amoxicillin/clavulanate suggests that 12.4% of the strains were β -lactamase producers. As expected, a larger percentage (93.8%) of the *M. catarrhalis* strains were producers of β -lactamase. The β -lactamase-stable compounds, including cefepime, cefotaxime, and cefpodoxime, were active against all of the *H. influenzae* and *M. catarrhalis* strains (100.0% susceptible). Resistance to the combination TMP-SMZ in these organisms is worrisome, because this drug is recommended for respiratory tract infections in the Essential Drugs Program of the WHO.⁵⁸

Table 2. SUSCEPTIBILITY OF RESPIRATORY TRACT PATHOGENS IN LATIN AMERICA—DATA FROM THE SENTRY ANTIMICROBIAL SURVEILLANCE PROGRAM, 1998*

Antimicrobial Agents	% Susceptible		
	<i>S. pneumoniae</i> (234)	<i>H. influenza</i> (242)	<i>M. catarrhalis</i> (96)
Penicillin	71.4†	—	—
Amoxicillin	85.5	87.2	6.2
Amoxic/Clavul	85.5	99.6	100.0
Cefuroxime	81.2	98.8	99.0
Cefpodoxime	—	100.0	100.0
Cefotaxime	88.9	100.0	100.0
Cefepime	86.8	100.0	100.0
Azithromycin	87.2	100.0	100.0
Clarithromycin	86.3	91.7	100.0
Chloramphenicol	94.0	98.3	100.0
Trimethoprim/ sulfamethoxazole	58.1	60.3	96.9
Tetracycline	74.4	98.3	96.9
Levofloxacin	100.0	100.0	100.0
Gatifloxacin	100.0	100.0	100.0

*572 isolates collected in 1998 in 10 centers in the following countries: Argentina, Brazil, Chile, Colombia, Mexico, and Venezuela

†The difference between the percentages of susceptibility to penicillin and ampicillin is caused by the differences in the break points (National Committee for Control of Laboratory Standards [NCCLS], 1999). The in vitro activity of these compounds is very similar against pneumococci.

Resistance in Organisms that Cause Enteric Infections

Antibiotics are widely prescribed in children with diarrheal disease, even though the cornerstone of treatment is the maintenance of adequate fluid-electrolyte balance.⁴² Indications for antibiotics are limited to very specific situations.⁴² Their inappropriate use is an important factor in the development of resistance by key pathogens, including *Shigella*, *E. coli*, and *Vibrio*. Bartoloni, from Bolivia, recently reported that 92% of adults and 40% of children received inappropriate treatment with antibiotics for acute diarrhea.¹

The program of the PAHO on intestinal pathogens³⁶ is currently active in Argentina, Brazil, Chile, Colombia, Costa Rica, Mexico, Peru, and Venezuela. Selected data from published results from the study are presented in Table 3.

Prescription practices based on trimethoprim/sulfamethoxazole or ampicillin need to be reevaluated in view of these results.

Antibacterial Resistance in Isolates from Community-Acquired Urinary Tract Infections

The bacterial causes of community-acquired urinary tract infections (UTI) (CAUTI) differ remarkably from that of UTIs occurring in hospital-

Table 3. PERCENTAGE OF RESISTANCE TO ANTIMICROBIAL IN SELECTED INTESTINAL PATHOGENS FROM LATIN AMERICA

	CHL	AMP	TMP/SMZ	FUR	CIP	TET
<i>Shigella flexneri</i>						
Chile	52	87	76	4	0	73
Colombia	80	40	33	—	—	—
Cuba	80	98	85	—	—	75
Venezuela	66	78	90	—	—	—
<i>Shigella sonnei</i>						
Chile	81	91	21	1	0	75
Venezuela	66	78	90	—	—	—

Data from references 13, 29, 47, 58

CHL = chloramphenicol; AMP = ampicillin; TMP-SMX = trimethoprim sulfa; CIP = ciprofloxacin; TET = tetracycline

ized patients, and the latter is frequently related to the presence of an indwelling urethral catheter. The causes of CAUTIs also differ in complicated infections because of altered renal function, abnormalities of the urinary tract, or underlying disease and in noncomplicated infections. The causes also differ according to localization of infection.

The commonest clinical presentation of CAUTI in girls and women is cystitis, and *Escherichia coli* is by far (in Latin America, as in the rest of the world) the most frequent causative agent; most of the remaining cases are caused by *Staphylococcus saprophyticus*. *E. coli* is also prevalent in pyelonephritis, but *Proteus mirabilis* (particularly in cases of struvite lithiasis) and *Klebsiella* spp are also involved. Complicated CAUTI are frequently caused by previous use of instrumentation in urologic procedures; *Pseudomonas aeruginosa* is a frequent isolate in these cases. Enterococci are also responsible for CAUTI, but their isolation is most frequent in older patients.⁸

In our area, the most important issue regarding CAUTI is the extreme resistance of *E. coli* to aminopenicillins and TMP-SMZ. Resistance levels in the survey of the Antimicrobial Committee of the Panamerican Society for Infectious Diseases performed in 1997, which included 2250 community-acquired *E. coli* isolates from nine countries,³⁸ had median values for ampicillin of 68% (range 58% to 92%) and for TMP-SMZ of 42% (range 38% to 71%). It has been shown that MICs for aminopenicillins in some Latin America *E. coli* isolates can be as high as 10,000 or 15,000 mg/L because of overproduction of TEM-1 β -lactamase and, in many instances, constitutive ampC overproduction.^{8,9}

Nitrofurantoin is a valid alternative for treating CAUTI, particularly in children. The previously mentioned API survey showed a median resistance value of only 4% in *E. coli* CAUTI isolates (range 2% to 9%). Because resistance to nitrofurantoin is not plasmid mediated, as it is to aminopenicillin and TMP-SMZ, practically no changes in resistance to nitrofurantoin have occurred since its introduction in the market 40 years ago. Problems with nitrofurantoin are lack of compliance because of bad taste (lower with macrocrystalline formulations), infrequent cases

of pneumonitis, and importantly, absolute resistance in *P. mirabilis* isolates.^{9, 31, 32}

Resistance to fluoroquinolones of *E. coli* has experienced a dramatic shift in Latin America. In the first 1993 API survey, only 1% to 3% of community-acquired isolates were resistant to ciprofloxacin. In the last surveillance survey from 1997, 7.5% of isolates were resistant (range 3% to 14%).

E. coli strains from Colombia showed resistance of 11% in 1997 to ciprofloxacin.⁴⁶ In a publication by Sifuentes et al in Mexico,⁵³ the resistance rate to ciprofloxacin in community-acquired strains of *E. coli* was 25%. In 1997, 21% of 2093 strains of this organism collected in the Venezuelan Surveillance System in 1998 were also resistant.¹²

Suarez et al⁵⁴ gathered data from 10 hospitals in Venezuela in 1998 and found that ceftibuten and cefixime were the most active oral drugs against urinary isolates of *E. coli*.

Extended-Spectrum β -Lactamases in Latin America

The extended-spectrum β -lactamase (ESBL) genes result generally from point mutations in the genes of broad-spectrum β -lactamase Ambler class A enzymes, such as TEM-1, TEM-2, or SHV-1, and are usually located in conjugative megaplasms, which often carry the genes responsible for resistance to other antibacterial drugs, causing serious difficulties for the treatment of infections caused by bacterial isolates producing those enzymes.²⁷

Until 1987, there were no reports of ESBLs from the southern cone of South America. That year, French investigators published the identification of the β -lactamase of a *Klebsiella pneumoniae* isolated in 1985 by Maria Eugenia Pinto in Santiago, Chile, as a new SHV derivative enzyme, and they named it SHV-5.²¹

In mid-1987, outbreaks of *K. pneumoniae* infections in a neonatal ICU that did not respond to treatment with the combination cefotaxime and amikacin were reported in Buenos Aires by Casellas et al.^{10, 11} The isolates were later confirmed as SHV-2 and SHV-5 producers. Further studies by the same investigators in the ICUs of San Lucas Hospital in San Isidro, Buenos Aires, showed the detection of putative ESBL producers in *Klebsiella* spp isolated from 1988 to 1989. Thirty-three of the 256 (14.5%) isolates were proven to produce SHV-2 or SHV-5 enzymes.^{7, 10, 11, 30}

A *K. pneumoniae* isolate kept lyophilized since 1982 during a preclinical study of cefotaxime in Buenos Aires also proved to be a producer of an SHV-5 ESBL.¹¹ We assumed that the selection of SHV point mutant enzymes happened as soon as third-generation cephalosporins were used in this area, probably because *K. pneumoniae* was at that time prevalent in Argentinian hospitals because of failures in infection-control measures.⁷

An outbreak involving 12 of 14 Argentinian provinces and cities of Bolivia and Paraguay caused by a multiresistant *Salmonella typhimurium*

that caused severe infections, including meningitis and sepsis in more than 300 children, occurred in late 1989.^{30, 33} One strain was identified as an ESBL producer, with pl 8.0 Bauerfeind et al identified a new ESBL and named it CTX-M-2,³ and subsequently it was sequenced.⁴ CTX-M-2 is a member of the CTX-M family, also known as cefotaximases. It exhibits higher activity with cefotaxime than with ceftazidime.

Currently in Argentina, the approximate distribution of ESBLs is as follows¹⁸: CTX-M-2, 64%; SHV-5, 6%; PER-2, 5%; SHV-2, 5%; SHV-4, 1%; others are unknown, but not related to the TEM family. CTX-M-2 has been detected so far in *K. pneumoniae* (80%). *E. coli*, *P. mirabilis*, *Shigella* spp, *Salmonella* spp, *Morganella morganii*, *C. freundii*, *Serratia marcescens*, *Serratia liquefaciens*, and *V. cholerae* El Tor. PER-2 has been detected in *P. mirabilis*, *K. pneumoniae*, *E. coli*, *Salmonella* spp, *E. cloacae*, and *V. cholerae* El Tor.^{19, 41, 43}

One *K. pneumoniae* strain with *ampC* inserted in a plasmid (FOX-1) was recently isolated in Argentina.²⁵ *Acinetobacter baumannii* with a plasmidic carbapenemase (ARI-2) was also recently isolated.⁵

No TEM-derived ESBLs have ever been found in Argentina, in spite of the huge number of *E. coli* strains carrying TEM-1 and overproducing it.⁴¹ Patterson's studies of the global presence of ESBLs³⁹ are in agreement with this observation.

The SENTRY Antimicrobial Surveillance Program is also evaluating the prevalence of ESBL-producing strains among Enterobacteriaceae (Table 4).

Using the screening concentrations recommended by the National Committee for Control of Laboratory Standards (NCCLS)³⁶ to predict isolates of *Klebsiella* spp and *E. coli* suspected of harboring an ESBL (MIC ≥ 2 μ g/mL to ceftazidime, aztreonam, or ceftriaxone), 8.5% to 8.9% of *E. coli* and 43.2% to 44.4% of *Klebsiella* spp strains isolated from bloodstream infections met these criteria. When we analyzed the data separated by country, the percentage of ESBL-producing strains causing blood strain infections (BSI) varied from 4.5% (Uruguay) to 12.0% (Chile and Mexico) among *E. coli*, and from 31.0% (Mexico) to 56.6% (Brazil) among *K. pneumoniae*. In a recent Brazilian study, approximately 40% of *K. pneu-*

Table 4. PREVALENCE OF ESBL-PRODUCING STRAINS IN LATIN AMERICA—DATA FROM THE SENTRY ANTIMICROBIAL RESISTANCE SURVEILLANCE PROGRAM, 1997*

Site	% ESBL (ESBL/total)	
	<i>E. coli</i>	<i>K. pneumoniae</i>
Bacteremia	8.9% (25/281)	44.4% (74/169)
Pneumonia	30.3% (10/33)	38.6% (22/57)
UTI	8.1% (20/246)	37.7% (23/60)
Wound	15.0% (9/60)	43.6% (17/39)
TOTAL	10.3% (64/620)	41.8% (136/325)

*Isolates from Argentina, Brazil, Chile, Colombia, Mexico, and Uruguay collected in 1997 and tested by broth microdilution (NCCLS) at the University of Iowa

moniae isolates collected in a São Paulo University hospital (Hospital São Paulo, Federal University of São Paulo) were ESBL producers. The majority of isolates from this study were from BSI.²⁰ In another study that evaluated isolates collected between December of 1995 and March of 1996 in Brazilian hospitals, only 87% of the *Klebsiella* isolates were susceptible to cefoxitin, suggesting that resistance to β -lactams may also be caused by mechanisms other than ESBL production (plasmid-mediated *ampC* or porin alterations).⁴⁸

In another recent surveillance study conducted in more than 35 Brazilian hospitals, we observed that 48% of *K. pneumoniae* and 10% of *E. coli* isolates were considered ESBL producers using the same criteria. This surveillance included different institutions, from small private clinics to large tertiary university medical centers.²²

Nonfermentative Gram-Negative Bacilli: *Pseudomonas aeruginosa* and *Acinetobacter baumannii*

The nonfermentative gram-negative bacilli *Pseudomonas aeruginosa* and *Acinetobacter baumannii* have become important causes of nosocomial infections in recent years. Because of intrinsic antimicrobial resistance, these pathogens can survive and proliferate under heavy antibiotic selective pressure. Results from the SENTRY Antimicrobial Surveillance Program indicated that *P. aeruginosa* was the most common species isolated from respiratory specimens collected from patients with nosocomial pneumonia in the Latin American hospitals evaluated (149 strains; 26.8%). In addition, it was the sixth most frequent cause of bacteremia and the third most frequent cause of urinary tract and wound infections. *A. baumannii* was the third most frequent pathogen isolated from patients with nosocomial pneumonia. These two pathogens were isolated from almost 40% of the patients with pneumonia. *A. baumannii* sp was also the seventh most frequent cause of bacteremia and urinary tract and wound infections. Both pathogens presented high rates of resistance to most antimicrobial agents evaluated in the SENTRY program.^{14, 47, 49, 50}

The length of hospitalization, previous use of third-generation cephalosporins, and concurrent chronic obstructive pulmonary disease have been described as the most significant predictors of colonization and infection in patients with nosocomial pneumonia caused by *P. aeruginosa*.⁵⁶ An increase in the risk of death has also been reported in these cases.⁴⁵ The antimicrobial susceptibility patterns of this pathogen in Latin America are shown in Table 5. In general, the most active compound against *P. aeruginosa* was the carbapenem meropenem (77.2% to 94.1% susceptibility). Piperacillin/tazobactam was the second most active compound (71.8% to 84.3% susceptibility), followed by amikacin (55.2% to 86.3% susceptibility) and imipenem (65.0% to 80.4% susceptibility). Cefepime (50.5% to 70.6% susceptibility) and ceftazidime (54.3% to 70.6% susceptibility) showed similar in vitro activity against this pathogen.

Isolates from urinary tract infections showed the highest rates of

Table 5. ACTIVITY OF ANTIMICROBIAL AGENTS AGAINST *P. AERUGINOSA* AND *A. BAUMANNII* ISOLATES CAUSING NOSOCOMIAL INFECTIONS IN LATIN AMERICAN HOSPITALS—DATA FROM THE SENTRY ANTIMICROBIAL SURVEILLANCE PROGRAM, 1997*

Antimicrobial Agent	% Susceptible Strains	
	<i>P. aeruginosa</i> (n = 271)	<i>A. baumannii</i> (n = 80)
Imipenem	72.3	91.3
Piperacillin / tazobactam	77.1	18.8
Cefepime	64.6	28.8
Ceftazidime	63.5	20.0
Ciprofloxacin	59.8	18.8
Amikacin	72.0	30.0
Gentamicin	56.8	—
Tobramycin	—	38.8

*Includes isolates from bloodstream infections, pneumonia, and urinary tract infections collected in hospitals located in Argentina, Brazil, Chile, Colombia, Mexico, and Uruguay. Data from Sader HS, Jones RN, Gales AC, et al: Antimicrobial susceptibility of patterns for pathogens isolated from patients in Latin American medical centers with a diagnosis of pneumonia: Results from the SENTRY surveillance program (1997). *Diagn Microbiol Infect Dis* 32:289, 1998.

resistance. In general, resistance rates were much higher than those seen in the United States and Canada.^{14, 40}

Acinetobacter spp were the second most frequent gram-negative bacilli isolated from patients with nosocomial pneumonia. *A. baumannii* is a pathogen of considerable importance because of the high mortality related to hospital-acquired pneumonia patients and the fact that it is usually resistant to multiple antimicrobial agents.^{2, 17} In our study, the carbapenems were active against 89.0% of *Acinetobacter* spp isolates (MIC₉₀, 8 µg/mL for imipenem and meropenem). Tetracyclines were also active in vitro. The next most active compound was tobramycin, which inhibited only 36.2% of the isolates tested (MIC₅₀, 16 µg/mL). Only the carbapenems represent viable options for treating *Acinetobacter* infections in the hospitals evaluated (see Table 5). These findings are of great concern. The high prevalence of multiresistant, nonfermentative gram-negative bacilli as a probable cause of respiratory tract infections would stimulate the use of carbapenems and, consequently, possibly greater carbapenem resistance in Latin American medical centers.

Staphylococci

Oxacillin resistance of staphylococci has been a chronic problem in most Latin American tertiary hospitals. Data from the SENTRY program have shown that 30% to 50% of *S. aureus* strains collected in Latin American hospitals are resistant to oxacillin and most antistaphylococci drugs available, except the glycopeptides (Table 6).^{40, 47} The newer fluor-quinolones, such as gatifloxacin and trovafloxacin, have demonstrated some activity against ciprofloxacin-resistant *S. aureus*; however, MICs

Table 6. ANTIMICROBIAL RESISTANCE RATES OF *S. AUREUS* ISOLATES FROM VARIOUS INFECTION SITES COLLECTED FOR THE SENTRY ANTIMICROBIAL SURVEILLANCE PROGRAM*

Antibiotic	% Resistant Strains		
	Bacteremia (n = 165)	Pneumonia (n = 127)	Wound (n = 174)
Oxacillin	30	50	31
Cefepime	30	50	31
Clindamycin	25	47	28
Trimethoprim/ sulfamethoxazole	22	40	20
Ciprofloxacin	26	45	29
Trovafoxacin†	10	5	4
Gatifloxacin†	2	3	3
Teicoplanin	0	0	0
Vancomycin	0	0	0

*Isolates from Argentina, Brazil, Chile, Colombia, Mexico, Uruguay, and Venezuela collected in 1997 (wound, 1997 and 1998) and tested by broth microdilution (NCCLS) at the University of Iowa

†Most ciprofloxacin-resistant isolates had intermediate MICs for the newer fluoroquinolones gatifloxacin and trovafoxacin.

are usually around the susceptibility break point, and most isolates are categorized as intermediately resistant to these newer compounds.^{40, 47} To date, no confirmed reports of *S. aureus* strains with reduced susceptibility to glycopeptides have been made in Latin America.

Vancomycin-Resistant Enterococci

Glycopeptide-resistant enterococci, best-known as vancomycin-resistant enterococci (VRE), have emerged as an important human pathogen responsible for serious systemic infections, especially in debilitated hosts with lowered defense mechanisms.¹⁶ Infection or colonization by VRE is more common in teaching hospitals and larger hospitals. Risk factors for acquisition of VRE include previous infection or colonization by multiresistant enterococci, severe underlying disease, previous prolonged hospitalization, and prior use of multiple antibiotics including vancomycin.¹⁵

There are scant data in the literature regarding VRE in Latin America; however, in spite of the continuing increase in the vancomycin resistance rates of enterococci in the United States, this pathogen has been rarely isolated in Latin American hospitals.^{13, 34} A recent multicenter study involving 15 Latin American medical centers in five countries (Argentina, Brazil, Chile, Ecuador, and Venezuela) collected only 24 VRE isolates (21 *Enterococcus faecium* and three *E. faecalis*) (Table 7). The prevalence of VRE could not be determined because of the design of the study; however, the study demonstrated an interesting fact: the majority of vancomycin-resistant *E. faecium* isolates were also resistant to the new streptogramin quinupristin/dalfopristin, which has not been used in

Table 7. ANTIMICROBIAL SUSCEPTIBILITY OF 429 ENTEROCOCCI STRAINS ISOLATED IN SOUTH AMERICA—RESULTS FROM THE GLOBAL SYNERCID* MICROBIOLOGICAL ASSESSMENT OF RESISTANCE TRENDS (GSMART, 1998–1999)

Organisms/ Antimicrobial Agents	MIC (μg/mL)		% Susceptible/ % Resistant by*	
	50%	90%	Brazil	Other Countries†
Vancomycin-susceptible				
<i>E. faecium</i> (94)			(n = 49)	(n = 45)
Quinopristin/dalfopristin	0.5	2	80/2	96/2
Ampicillin	0.5	16	84/16	72/28
Vancomycin	0.5	2	100/0	100/0
Teicoplanin	0.25	0.5	100/0	100/0
Ciprofloxacin	2	>2	33/53	24/24
Chloramphenicol	4	16	88/8	87/0
Doxycycline	2	>8	78/16	61/22
Vancomycin-resistant				
<i>E. faecium</i> (21)			(n = 17)	(n = 4)
Quinopristin/dalfopristin	8	8	12/88	100/0
Ampicillin	>16	>16	0/100	0/100
Vancomycin	>16	>16	0/100	0/100
Teicoplanin	>16	>16	0/100	0/100
Ciprofloxacin	>2	>2	0/94	0/50
Chloramphenicol	8	8	94/6	75/25
Doxycycline	2	8	82/0	75/25
<i>E. faecalis</i> (223)			(n = 83)	(n = 140)
Quinopristin/dalfopristin	8	>16	1/90	13/71
Ampicillin	1	2	99/1	93/7
Vancomycin	1	2	98/2	96/1
Teicoplanin	0.12	0.12	98/2	100/0
Ciprofloxacin	2	>2	44/32	14/28
Chloramphenicol	8	>16	59/35	69/14
Doxycycline	4	>8	60/23	44/39
<i>Enterococcus</i> spp (91)			(n = 44)	(n = 47)
Quinopristin/dalfopristin	0.12	2	73/2	94/4
Ampicillin	0.50	2	91/9	89/11
Vancomycin	0.25	4	91/9	87/0
Teicoplanin	0.12	0.25	91/9	94/2
Ciprofloxacin	1	>2	72/14	11/43
Chloramphenicol	4	8	91/9	36/30
Doxycycline	2	2	98/2	23/60

*Quinopristin/dalfopristin (Rhône Poulenc) susceptibility defined as MIC ≤ 1 μg/mL or ≥ 19 mm.

†Isolates from Brazil (five medical centers) were tested by broth microdilution and isolates from the other countries (Argentina, Chile, Ecuador, and Venezuela—10 medical centers) were tested by disk diffusion.

Latin America.⁵¹ Streptogramin resistance was also detected in vancomycin-susceptible strains.

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Address reprint requests to

Manuel Guzmán-Blanco, MD
 Consultorio 37 anexo A
 Centro Médico de Caracas
 Plaza El Estanque, San Bernardino
 Caracas, Venezuela 1010